Emerson Cortazar

914-619-8529 | efc19@duke.edu | linkedin.com/in/emerson-cortazar | github.com/emersoncortazar

Research Interests

Robot learning and general-purpose robot policies, AI agents for physical systems, robot perception and SLAM, and field robotics for unstructured environments.

Education

Duke University Durham, NC
BSE in Electrical & Computer Engineering, Certificate in Robotics & Automation *2024 – 2028*

- GPA: 3.81 / 4.00. Dean's List with Distinction, Fall 2025.
- Relevant coursework: Fundamentals of Electrical and Computer Engineering (ECE 110L), Signals and Systems (ECE 280L), Computer Architecture (ECE 250D), Data Structures and Algorithms (CS 201), Matrices and Vectors (MATH 218D-2), Multivariable Calculus (MATH 219).

New York University, Tandon School of Engineering Brooklyn, NY
Visiting student through Duke in New York *Fall 2026*

- Coursework in progress: Combinatorics (MA-UY 4314).

Research Experience

NYU AI4CE Robotics Lab Brooklyn, NY
Undergraduate Research Assistant *Sep 2026 – Present*

- Contribute to Agentic SLAM, a project in which an AI model monitors how well a robot's camera-based localization (ORB-SLAM3) is tracking and adjusts its settings when tracking starts to fail.
- Ran an experiment testing which inputs (camera frames, trajectory plot, tracking telemetry) let a vision-language model tell healthy ORB-SLAM3 runs from degraded ones on a TUM-VI sequence.
- Traced the model's false alarms on healthy runs to a time-span mismatch between the trajectory plot and the camera frames; matching the two cut false alarms from 8 of 9 trials to 1 of 9.
- Presented the experiment and its findings at lab meeting; extending the analysis to trajectory data.

Duke Polarization Lab Durham, NC
Undergraduate Research Assistant *Oct 2025 – May 2026*

- Compared several large language models on how closely their generated text resembles comments written by people on Twitter and Reddit, across a set of similarity tests.

Industry Experience

Hyde Park Partners *Seasonal Engineering Intern*
Industrial technology company. Returned across multiple terms in progressively technical roles at three subsidiaries.

Automation Solutions, Inc. Birmingham, AL
Robotics & Controls *Summer 2026*

- Designed an automated part-picking cell for Toyota: configured the 3D machine-vision system to reliably detect parts in bins, laid out the cell's safety perimeter, and programmed the robot arm's motion onto production conveyors.
- Modified control logic and designed the operator HMI screen for a heavy-lift platform built for a SpaceX facility.

Livingston & Haven Charlotte, NC
Electrical Engineering *Summer 2026*

- Assembly team member building a 1.2-megawatt microgrid for large manufacturers and small-scale data centers.
- Built industrial control panels for hydraulic power units that operate heavy manufacturing equipment.

AEG International Charlotte, NC
Electrical Engineering R&D *Summer 2025*

- Prototyped and wrote manuals for solar microgrids, solar streetlights, and water filtration devices at a clean-energy solutions company.
- Produced a complete solar microgrid system design and pricing proposal for a prospective commercial customer evaluating the product for their factory.

Technical Projects and Leadership

Duke Applied Machine Learning Group Durham, NC
Project Manager *2025 – Present*

- Manage a team of three other undergraduates building a parts catalog enrichment and recommendation system for MRO Stop, an industrial parts distributor, working directly with the company's CEO and chairman of the board.
- The system extracts specifications from manufacturer pages to fill in a 292,000-row catalog and recommends equivalent parts by combining text similarity (TF-IDF) with attribute matching (Python, pandas, FAISS).

DukeEngage Eswatini Hhohho, Eswatini
Cultural Relations Manager and electrical system designer *Summer 2025*

- Spent a Duke-funded service term with the nonprofit Engineers in Action building a solar-powered clean water system for a rural school of 300 students.
- Designed and built the electrical system for the solar-powered water system as the team's electrical engineering student.
- Led community events, youth sanitation classes, and technical presentations on the system for teachers and local community members.

Teaching

Duke Ignite Learners Durham, NC
Teacher and Family Communication Manager *Sep 2024 – Present*

- Teach a class of 20 middle schoolers in a paid eight-week program that supports Dr. Megan Madonna's research on effective STEM teaching methods, guiding students from basic circuits through building their own working light sources.
- Serve as the program's point of contact for families, sending weekly progress updates and handling parent questions.

Activities

- **Alexander Hamilton Society, Duke chapter.** Publicity Chair, 2025 – 2026. Ran the chapter's email list and newsletter and handled logistics for meetings with visiting think tank speakers.
- **Phi Alpha Delta Pre-Law Society.** Philanthropy Committee, raising funds for local legal support in Durham.
- **Duke Club Tennis** and **Duke Club Lacrosse.**

Skills

- **Programming:** Python, C++, C, Java, R, HTML
- **Libraries:** NumPy, pandas, PyTorch, OpenCV, FAISS
- **Robotics and controls:** FANUC RoboGuide, iRVision, Siemens TIA Portal
- **Tools:** Linux, Git, MATLAB, AutoCAD Electrical, Fusion 360
- **Languages:** English (native), Spanish (basic)